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Roll No

EE-7201 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
High Voltage Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Explain the streamer theory of breakdown in air at atmospheric pressure.
2. What is Paschen's law? How do you account for the minimum voltage for breakdown under a given 'p×d' condition.
3. What do you understand by "intrinsic strength" of solid dielectric? How does breakdown occur due to electrons in a solid dielectric?
4. Describe the Marx circuit arrangement for multistage impulse generators. How is the basic arrangement modified to accommodate the wave time control resistances?
5. Explain the different breakdown mechanisms occurs in solid dielectrics in practice?
6. Explain the procedure for performing.
 - i) IR test
 - ii) Stability test
7. Define Townsend's first and second ionization coefficients. How is the condition for breakdown obtained in a Townsend discharge?

OR

What is the use of Tesla coil?

8. Explain the following (Any two) :
 - a) Need for generating high voltages in laboratory.
 - b) Important applications of high voltage.
 - c) BIL.
 - d) Methods of insulation co-ordinator.
